售后技术中心

Xiaomi Tablet 4 Plus Level 3 Maintenance Guide V01

Technical support internal document control: TSIMHND9P Xiaomi Tablet 4 Plus Level 3 Maintenance Guidance V01

Scope of application:

Analysis center, mainboard and whole machine maintenance factory

Change history:

First edition 2018-11-20

1. Introduction of basic information

1.1 Product Overview

Product Overview:

处理器与容量	骁龙 660 AIE 处理器，最高主频 2.2GHz Adreno 512图形处理器，最高主频 650MHz 小米平板4 Plus 有4GB+64GB/4GB_128GB 两个版本可选 LPDDR4x 双通道内存，Emmc 5.1 高速闪存
外观尺寸	宽：149.08mm 高：245.6mm 厚：7.99mm 重：485g
续航	8620mAh（typ）大电量，支持 5V/2A 充电，标配 5V/2A 充电器
屏幕	10.1 英寸，16:10 屏幕比例，1920*1200 FHD、224PPI、1000:1对比度 支持护眼模式，标准模式，增强模式，夜光屏，智能适应模式下无极色温
拍照与摄像	1300万像素后置相机:f/2.0光圈、支持HDR、支持美颜、支持 1080p 视频拍摄 500万像素前置相机：f/2.0光圈、支持前置美颜、支持 HDR、支持人脸识别
网络	WiFi+4G LTE版，支持单卡不限运营商，Nano-SIM卡槽，支持移动/联通/电信 4G， 支持移动/联通4G+ 支持频段 TDD-LTE:B34/B38/B39/B40/B41 FDD_LTE:B1/B3/B5/B7/B8
无线网络	支持 802.11a/b/g/n/ac 支持 2.4G WiFi/支持 5G WiFi/支持 WiFi Direct 支持 2*2Wi-Fi MIMO 支持 蓝牙5.0
导航与定位	4G LTE版：GPS AGPS GLONASS Beidou
传感器	4G LTE版：陀螺仪 加速感应器 环境光传感器 霍尔传感器 电子罗盘
多媒体播放	音频格式：ACC LC;HE-AACv1(ACC+);HE-AACv2(ACC+);ACC;AMR-NB;AMR-WB;FLAC; MP3;PCM/WAVE;Vorbis 视频格式：H.263;H.264 AVC Baseline Profile(BP);H.264 AVC Main profile(MP); H.264 HEVC;MPEG-4 SP;XVID;VP8;VP9;ASF/WMV;3G2;AVI;TS;MKV/WEBM;M4V;3GP/MP4; FLV

1.2 Special welding fixture for millet flat plate 4 Plus

Material code: SCNC030044400

1.3 Xiaomi flat plate 4 Plus power supply adapter

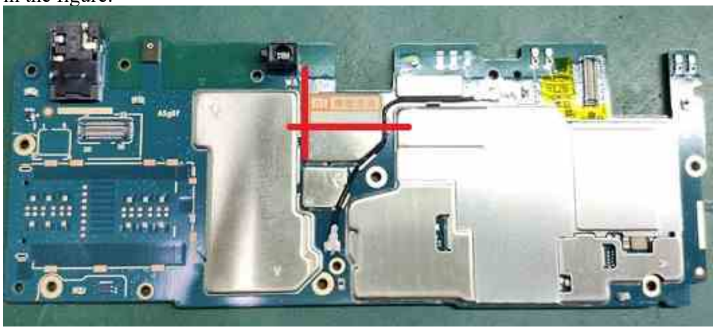
Material code: SCNC030043000

1.4 Millet flat plate 4 Plus RF test jig

Material code: SCNC030044500

1.5 Pasting position and specification of maintenance label

Xiaomi Tablet 4 Plus Maintenance Label: The pasting position is as shown in the figure:



The service label is 12mm long * 5mm wide, and is pasted on the upper left corner of SH2101 shielding frame.

1.6 Matters needing attention in motherboard maintenance

Note:

1. There is a binding relationship between CPU and EMMC. EMMC must be replaced at the same time when replacing CPU, and EMMC can be replaced separately. However, after replacing EMMC, it can be turned on normally in factory mode. If TEEKEY is not uploaded, it will be turned on and restarted in user mode.
2. When maintaining and heating the motherboard, it is necessary to take off the shielding frame of CPU shielding shell, so as to avoid other faults caused by the expansion and extrusion of heat-conducting silica gel after overheating.
3. Take anti-welding measures when welding the components near the key interface, display interface and earphone interface. These plastic components are easy to weld.
4. Enter CIT mode in user mode, click "Settings-All Parameters-Kernel Version", and click "Kernel Version" four times in succession to enter CIT test mode.

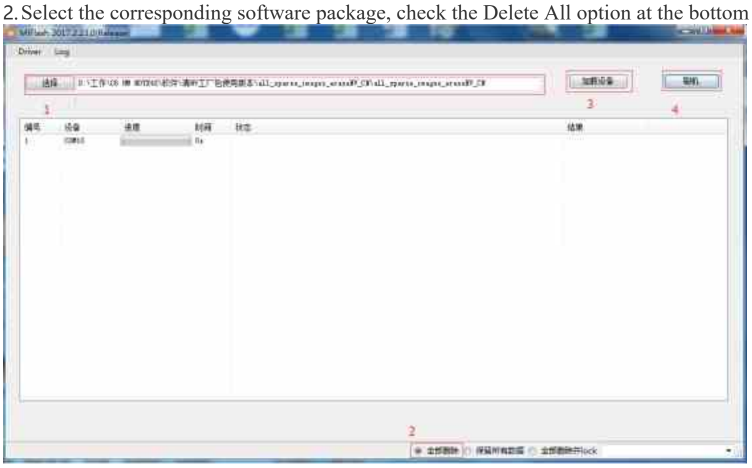
1.7 Brush mode

Brush platform: MiFlash

Special factory package for changing number (keep NV package): Clover-V017-keeper NV factory package for maintenance (clear NV package): Clover-V017-eraseNV factory package Brushing machine mode: deep brushing machine

1. When the mobile phone is turned off, short-circuit TP2103 and TP2104 to make the mobile phone enter deep brush

2. Select the corresponding software package, check the Delete All option at the bottom of MiFlash, click Load Device, and click Brush Machine.



1.8 RF calibration test correlation

1. Deep brush factory software, write FSN with DT tool.

2. Open the calibration software, and connect the main receiving antenna and the auxiliary receiving two RF lines when the mobile phone is turned on. Connect the USB cable and the RF calibration of the mobile phone starts (refer to the calibration operation instruction (XTT) of D9P_LTE version for details).

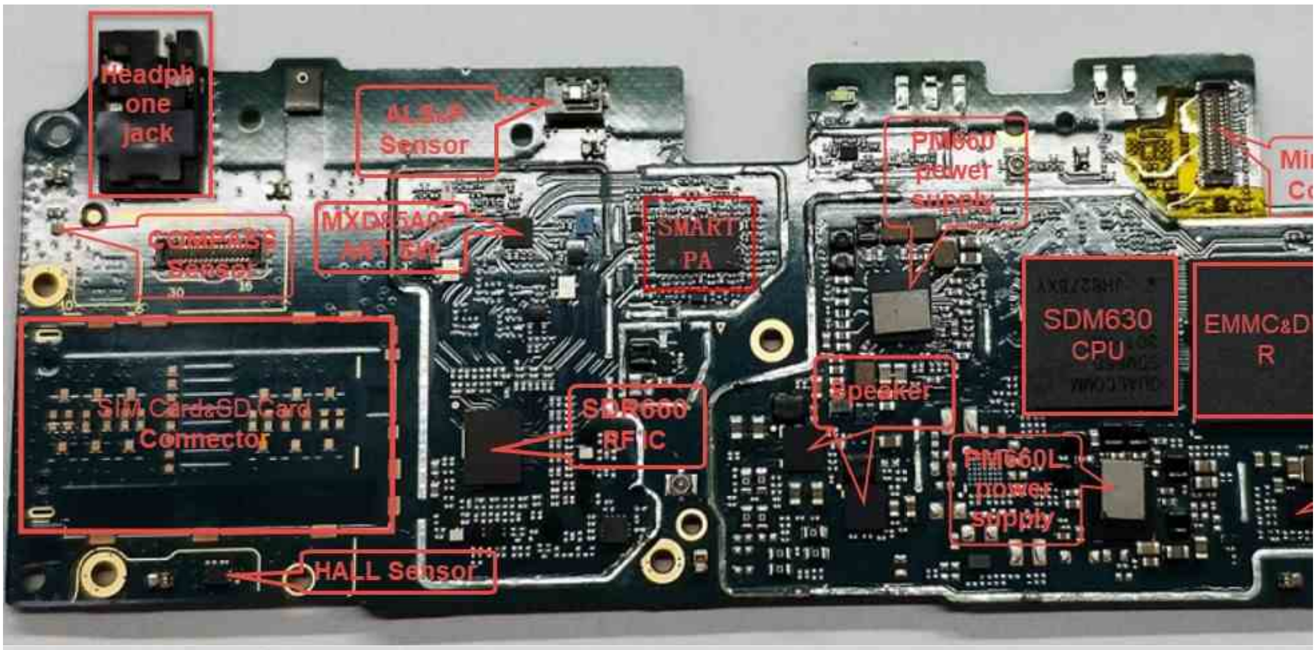
3. After successful calibration, use the "NV Read & Write V2.2 Tool" to write the flag bit (refer to the NV Read & Write V2.2 Operation Instructions for details).

4. After restarting with DT tool, you can directly insert the card to test the signal.

2. Brief introduction of motherboard module

2.1 Component distribution map of Xiaomi tablet 4 Plus motherboard

- TOP plane



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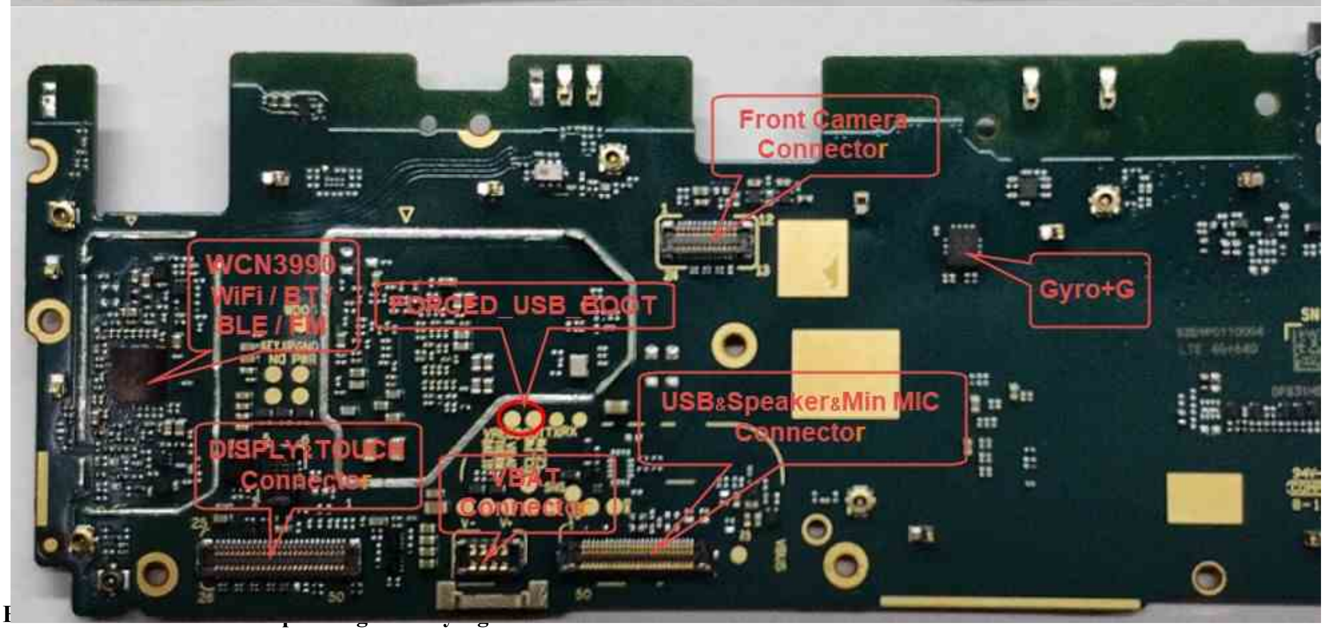


2.2

Boot Timing:

开机时序测量表		
Symbol	测量值	测量点
KYPDPWR_N	1.8V	TP2805
VREG_BOB	3.3V	C1403
VREG_S5B_0P915	0.915V	L1705
VREG_S3B_S4B_0P87	0.87V	L1703
VREG_S2B_1P05	1.05V	L1702
VREG_L3A_1P0_SDR_DVDD	1V	C1406
VREG_S4A_2P04	2.04V	C1806
VREG_S5A_1P35	1.35V	C1313
VREG_L13A_1P8	1.8V	C1415
VREG_L10A_1P8	1.8V	C1412
VREG_L9A_1P8	1.8V	C1411
VREG_L6A_1P3	1.3V	C1408
VREG_L11A_1P8	1.8V	C1413
REF_MSM_HVPAD	1.25V	C1042
VREG_L10B_0P915	0.915V	C1717
VREG_L9B_0P87	0.87V	C1716
VREG_L1B_0P925	0.925V	C1034
VREG_L7B_3P125	3.125V	C1022
VREG_L2B_SDC2	2.95V	C1011
VREG_L5B_2P95	2.95V	C1712
VREG_S1B_1P125	1.125V	L1701
VREG_L4B_2P95	1.95V	C1711
VREG_L8A_1P8	1.8V	C2008
SLEEP_CLK	32KHz	R1213
VREG_S1A_0P87	0.87V	L1301
VREG_L14A_1P8	1.8V	C1416
VREG_L1A_1P225	1.225V	C1404
LNBBCLK	19.2MHz	R1201
PON_RESET_N	1.8V	R509
SDM_PS_HOLD	1.8V	TP2113

ent points (yellow signals are also signals in startup timing):

CPU供电测量表		
Symbol	测量值	测量点
VREG_S5B_0P915	0.915V	L1705
VREG_L10B_0P915	0.915V	C1717
VREG_L10A_1P8	1.8V	C1412
REF_MSM_HVPAD	1.25V	C1042
VREG_L2B_SDC2	2.95V	C1011
VREG_S1A_0P87	0.87V	L1301
VREG_S3B_S4B_0P87	0.87V	L1703
VREG_L9B_0P87	0.87V	C1716
VREG_L1B_0P925	0.925V	C1034
VREG_L1A_1P225	1.225V	C1404
VREG_L7B_3P125	3.125V	C1022
VREG_L13A_1P8	1.8V	C1415
VREG_L15A_UIM1	C1021（SIM1插卡）	2.9V
VREG_L17A_UIM2	C1026（SIM2插卡）	2.9V
VREG_L6A_1P3	1.3V	C1023
VREG_L9A_1P8	1.8V	C1030
VREG_S6A	0.7V	C901
VREG_L5A_0P848	0.848V	C1032
VREG_S1B_1P125	1.125V	C1010
VREG_S2A_S3A_0P87	0.87V	L1302

U1201 chip bitmap:

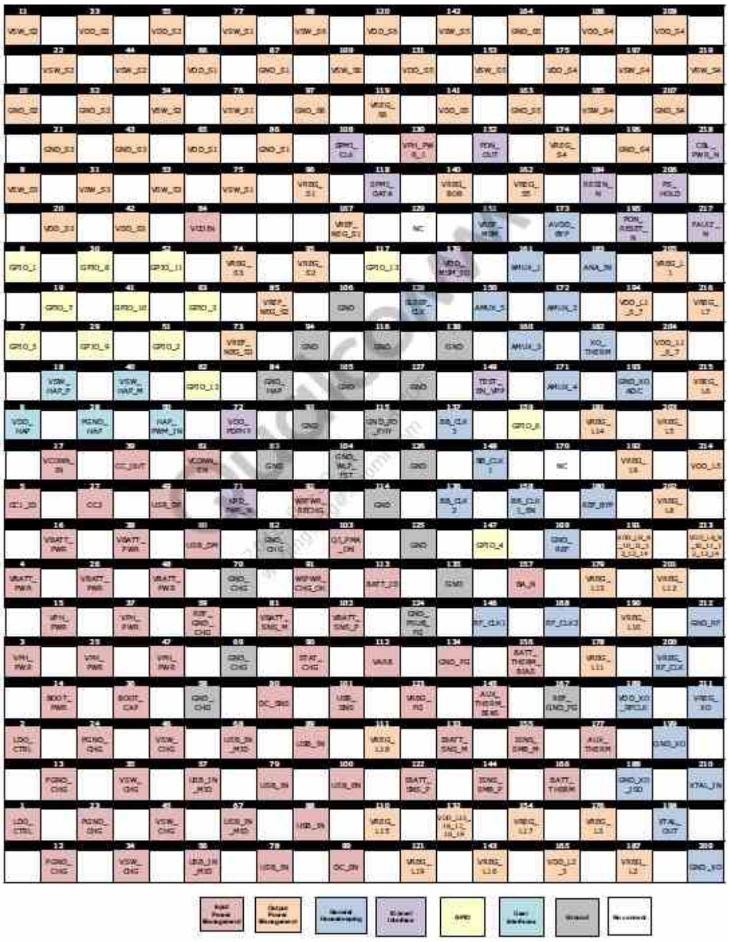


Figure 2-1 PM660 pin assignments (top view)

U1601 chip bitmap:

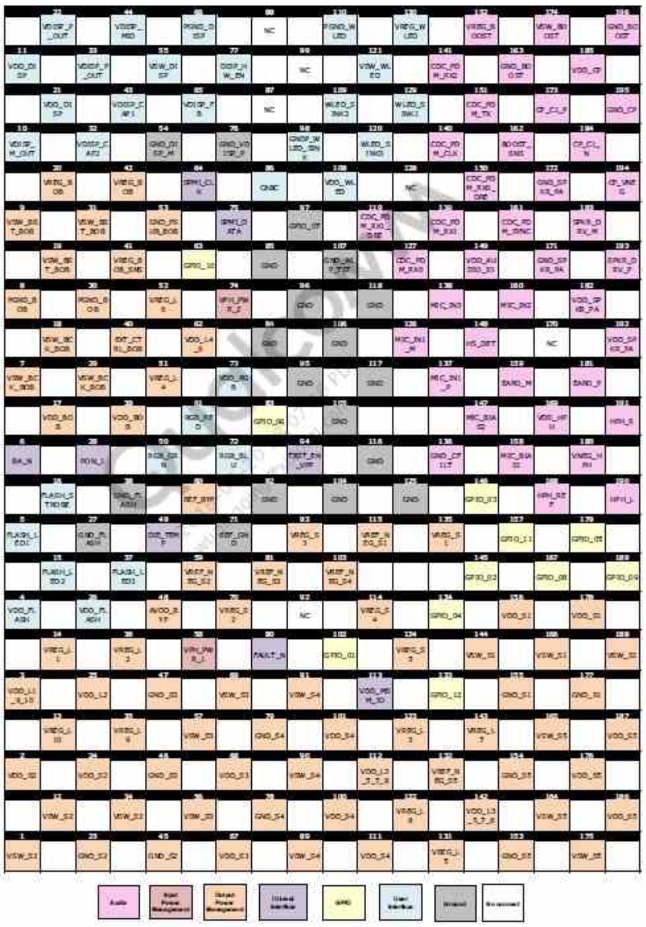


Figure 2-2 PM660L pin assignments (top view)

3. Troubleshooting

3.1 Switching failure

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. In specific measurement, we should measure according to the starting time sequence.

3.1.1 Constant current without startup

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. If the brushing machine reports an error, measure whether the working conditions of U2001 and U501 are normal (in actual maintenance, the brushing machine 11S reports an error "ACK count don't match" generally means that U501 is damaged).
3. If the current is still constant after brushing, measure whether the startup timing and CPU working conditions are normal.

Maintenance Case 1
Fault phenomenon: No startup,
120mA constant current fault cause:
U2001

Maintenance mode: No startup, 120 constant current, invalid brush machine, fault repair after replacing U2001.

Maintenance Case 2
Fault phenomenon: No startup,
122mA constant current fault cause:
U2001

Maintenance mode: No startup, 122mA constant current, error report by brushing machine, fault repair after replacing U2001.

Maintenance Case 3
Fault phenomenon: No startup,
195mA constant current fault
cause: U2001

Maintenance mode: Do not start up, 195mA constant current, brush the machine to report error, and repair the fault after replacing U2001.

3.1.2 No startup current is not maintained

Analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether the startup timing signal is normal.
3. Measure whether the power supply of U2001 is normal.
4. Measure whether the USB signal circuit is normal (BUS_USB_SNS; USB_HS_D_P_CP; USB_HS_D_M_CP; Note: If these measured signals are abnormal, CPU fault can be judged from the side).

Maintenance Case 1
Failure phenomenon: No boot, no
maintenance of 140mA Failure Cause:
Software upgrade

Maintenance mode: Do not start up, 140mA is not maintained, and the fault is repaired after brushing the machine.

Maintenance Case 2
Failure phenomenon: No startup,
70mA no maintenance Failure cause:
U501

Maintenance mode: No startup, 70mA is not maintained, no mouth is brushed, and the fault is repaired after replacing U501.

Maintenance Case 3
Fault phenomenon:
white rice power
failure fault
component: U501

Maintenance method: Power failure of white rice, invalid software upgrade, fault repair after replacing U501.

3.1.3 No startup, no current

Analysis ideas:

1. Check whether the appearance of J2803 is damaged, and if the interface is normal, measure whether the ground value of J2803 is normal (BAT_CON_ID, PCB_BATT_THERM, VBAT).
2. Power-on measurement VPH_PWR output is normal, KYPDPWR_N (TP2805) 1.8 V voltage is normal. If there is no output, replace U1201.
3. Measure whether the startup timing signal is normal.
4. Measure whether VPH and VBAT are short-circuited.

Maintenance Case 1
Fault Phenomenon: No power
on, no current Fault Cause:
C2529

Maintenance mode: No startup, no current when powered on, short circuit of VPH measurement, serious heating of C2529 when powered on, fault repair after removing and replacing C2529.

Maintenance Case 2

Fault phenomenon: No startup, no current

Cause of failure: C1303

Maintenance mode: No startup, power-on without current, short circuit of VPH measurement, serious heating of C1303 after power-on burning, and fault repair after removing and replacing C1303.

Maintenance Case 3
Fault phenomenon: C2211 machine is not opened, and there is no current fault reason:

Maintenance mode: No startup, power-on without current, short circuit of measuring VPH, serious heating of C2211, fault repair after removing and replacing C2211.

Maintenance Case 4
Fault Phenomenon: No power on, no current Fault Cause: C1721

Maintenance mode: No startup, power-on without current, short circuit of VPH measurement, serious heating of C1721, fault repair after removing and replacing C1721.

Maintenance Case 5
Fault Phenomenon: No power on, no current Fault Cause: C2824

Maintenance mode: No startup, no current when powered on, short circuit of measuring VBAT, serious heating of C2824 when powered on, and fault repair after removing and replacing C2824.

3.1.4 Leakage without starting up

Maintenance analysis ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.
2. Measure whether VBATT and VPH_PWR are short-circuited. If both VBATT and VPH_PWR are short-circuited, first find VBATT short-circuited elements, then find VPH_PWR short-circuited elements.
3. Measure whether there is short circuit in other power supply lines, and find out the faulty components according to the short circuit signal.
4. Power up to find heating elements.

Maintenance Case 1
Fault phenomenon: No startup, leakage 370mA Fault component: U1601

Maintenance mode: No startup, electric leakage 370mA, measured VPH resistance to ground is about 100 (normal about 400), heating is abnormal when burning U1601, and the fault is repaired after replacing U1601.

Maintenance Case 2
Fault phenomenon: No startup, leakage 260mA Fault component: U2501

Maintenance mode: No startup, leakage 260mA, measurement of LED_A voltage short circuit, replacement of U2501 after the fault repair.

3.2 Restart failure

Analysis ideas:

1. Check whether the 1.8 V voltage of KYPD_PWR_N signal is pulled low.
2. Software upgrade, troubleshooting software.
3. Use QDUTT tool to detect whether DDR is abnormal.
4. Measure whether I2C to ground value and voltage are normal.
5. Measure whether the clock is normal when U501 is powered.
6. Measure whether the motherboard has abnormal power supply caused by short circuit.
7. Replace U501.

Maintenance Case 1
Fault Phenomenon: White Rice Restart Fault Component: U2001

Maintenance mode: White rice is restarted, brush machine is invalid, and the fault is repaired after replacing U2001.

3.3 Crash failure

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Use QDUTT tool to detect whether DDR is abnormal.
3. Measure whether BAT_CON_ID and PCB_BATT_THERM are normal.
4. Measure whether the startup timing is normal.
5. Measure whether the power supply of U501 is normal.
6. Measure whether there is abnormal power supply caused by short circuit on the motherboard.
7. Replace U501.

Maintenance Case 1
Fault Phenomenon:

White Rice Fixed
Screen Fault
Component: Software
Upgrade

Maintenance mode: fix white rice when starting up, and repair the fault after brushing the machine.

3.4 Signal fault

Bitmap of Xiaomi Tablet 4 Plus RF Chip U3401:

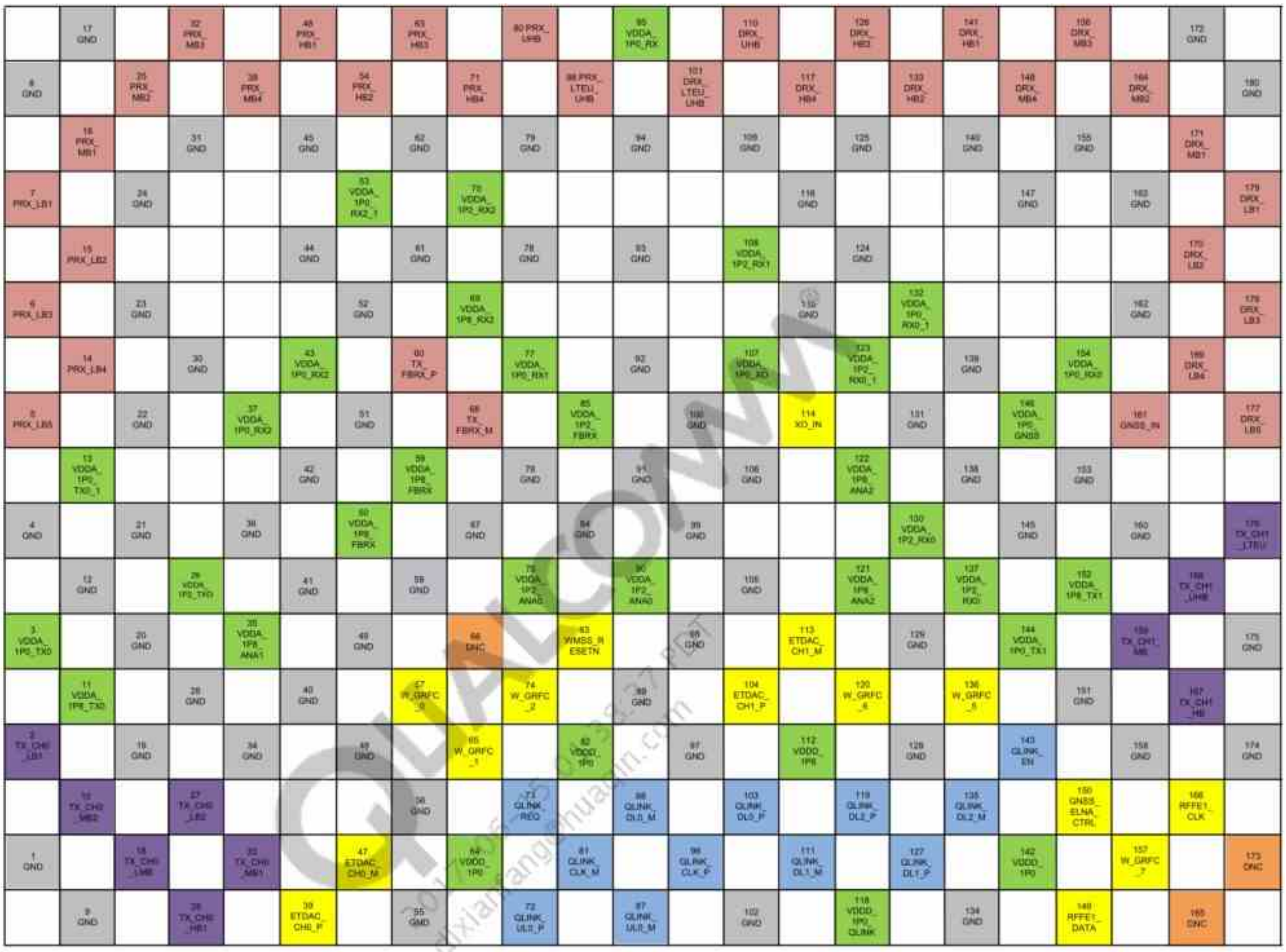


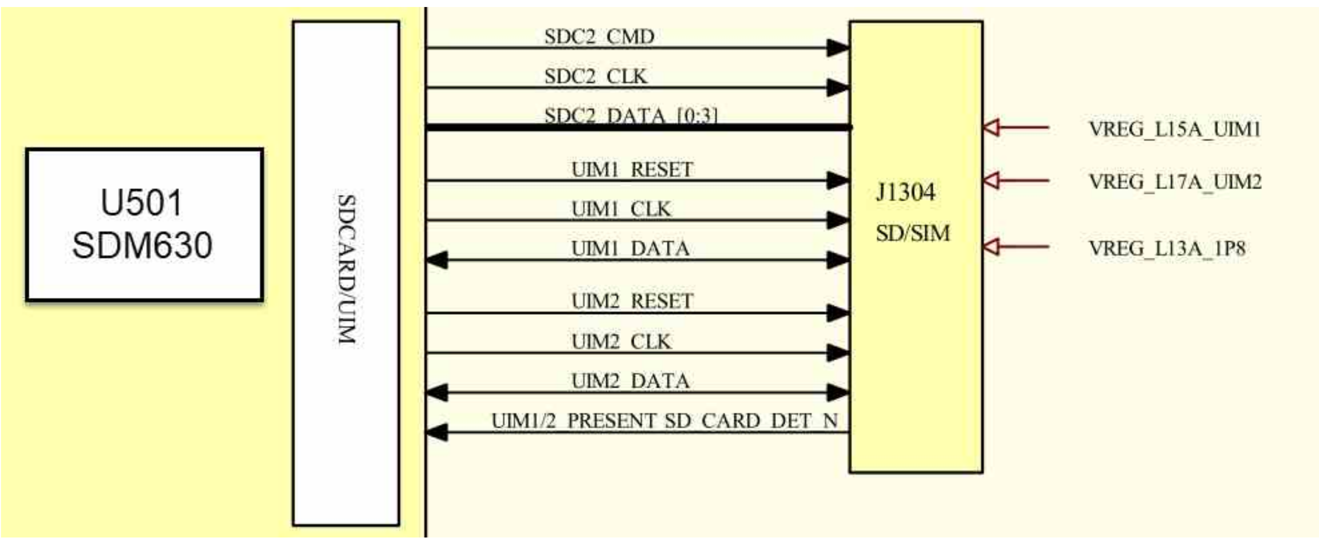
Figure 2-2 SDR660 pin assignments – top view

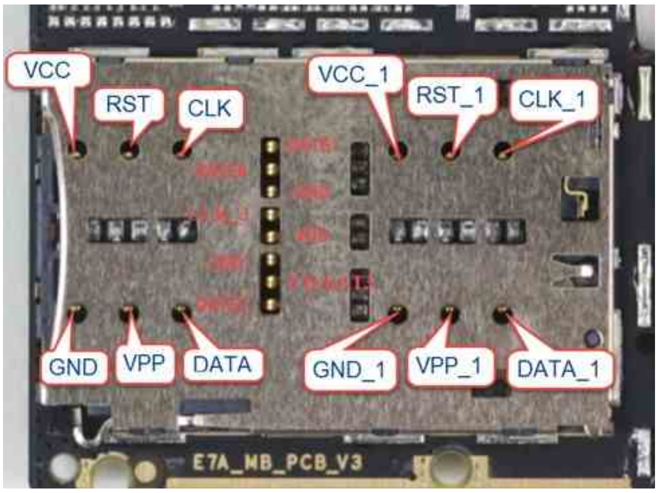
Analysis ideas:

1. Insert SIM card to ensure normal identification, and eliminate faults that do not recognize SIM card.
2. Software upgrade, troubleshooting software.
3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
4. Measure whether the power supply of RF circuit is normal and whether the RFFE CLK (1/2)/DAT1 signal is normal.
5. If the RF calibration is normal and there is still no signal, check the SIM card circuit and whether the J3802 to J3803 lines are normal.

3.5 SIM card failure

Principle block diagram of SIM card:





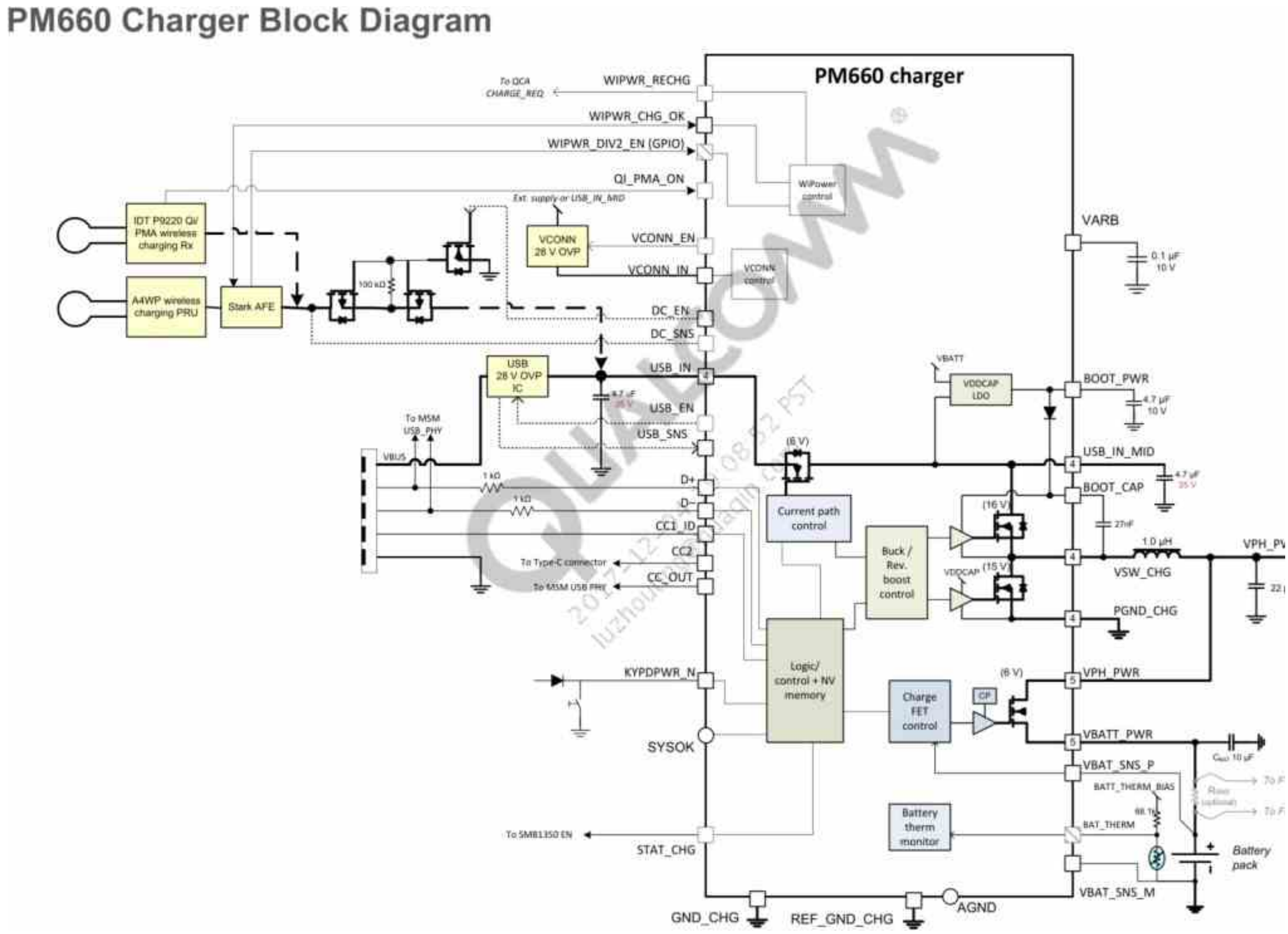
SIM/SD card meter:

SIM/SD测量表		
Symbol	测量值	测量点
VCC_1	650	SIM1
RST_1	430	
CLC_1	430	
VPP_1	1	
DATA_1	430	
VCC_2	650	SIM2
RST_2	430	
CLC_2	430	
VPP_2	1	
DATA_2	430	
DATE1	420	SD
DATE0	420	
CLK_3	439	
VDD	680	
CMD	420	
CD/DAT3	420	
DATE2	420	

1. First, check whether the baseband version of the mobile phone is normal. If the baseband information is normal, it is the SIM card related function failure.
2. Check whether the SIM card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joint of SIM card holder.
3. Measure whether the SIM card aims at the ground value normally.
4. Start up and test whether the power supply, clock and reset of SIM card are normal, and whether the data voltage jump is normal
5. If the above signal is normal, replace U501.

3.6 Charging failure

Charging schematic diagram:



Analysis ideas:

1. Check whether the appearance of J2802 and J2803 is normal, and measure whether the ground values of VBUS_USB_SNS, USB_HS_D_P_CON and USB_HS_D_M_CON are normal with multimeter diode gear.

2. Measure whether VBAT, AT_CON_ID and PCB_BATT_THERM of J2803 are normal to ground.

3. To ensure that the above signals are normal, consider replacing U1201.

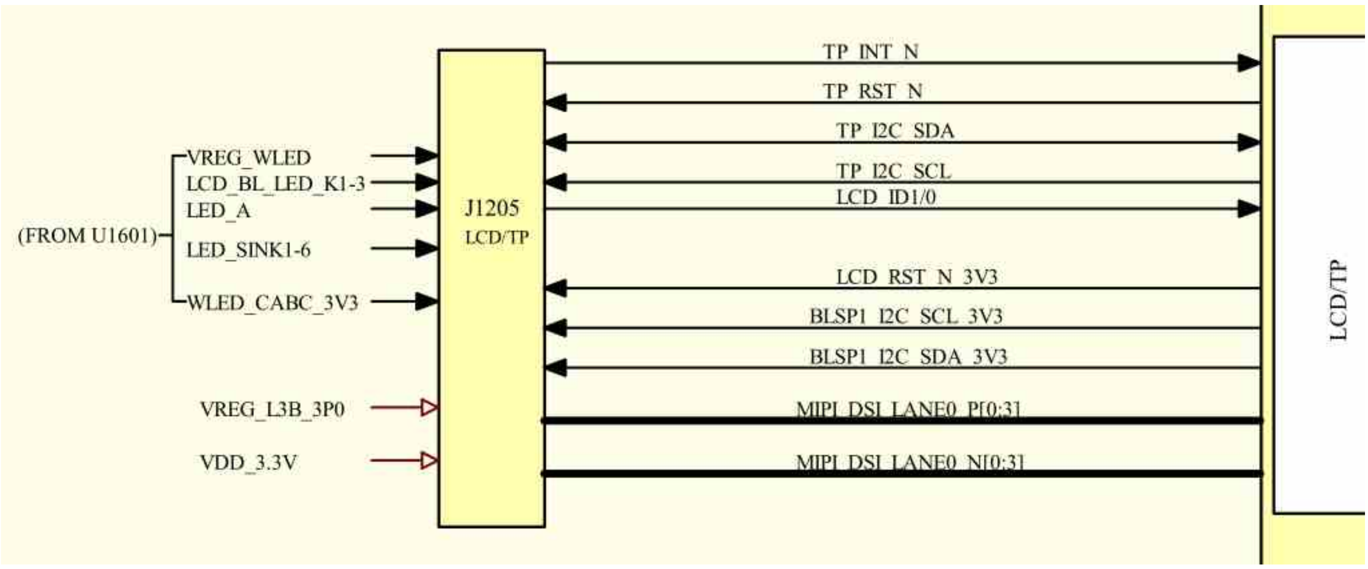
Maintenance Case 1

Fault Phenomenon:
No Charge Faulty
Component: D2824

Maintenance mode: USB_CC2_CON does not charge, measure TP2833 short circuit, and repair the fault after replacing D2824.

3.7 Display failure and touch screen failure

Principle block diagram:



Display and touch screen interface J2502 measurement:

GND	1	550	550	550	1	1	GND	410	410	410	410	1	532	532	532	532	1	596	596	596	1	1	1	GND
25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
J2502																								
26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
GND	260	260	GND	260	260	GND	260	260	GND	260	260	GND	260	260	GND	1	1100	990	990	1	600	600	600	GND

Display some measurement tables:

显示电压测量表		
Symbol	测量值	测量点
LDM_ID1	4V-29V	R2521
WLED_CABC_3V3	3.3V	R2535
LCD_RST_N_3V3	3.3V	R2510
VDD_3.3V	3.3V	C2527
BLSP1_I2C_SDA_3V3	3.3V	J2502第47脚
BLSP1_I2C_SCL_3V3	3.3V	J2502第48脚
LCD_BL_LED_K1	0.3V	R2501
LCD_BL_LED_K2	0.3V	R2502
LCD_BL_LED_K3	0.24V	R2517

Voltage measurement of touch screen part:

TOUCH电压测量表		
Symbol	测量值	测量点
TP_I2C_SDA	1.8V	R808
TP_I2C_SCL	1.8V	R809
TP_RST_N	1.8V	R2514
TP_INT_N	1.8V	R2513
VREG_L3B_3P	3.3V	R2509

Maintenance analysis ideas:

1. Visually check whether J2502 and its peripheral components are damaged or falsely welded.

2. Brush the machine to eliminate software faults.

3. Measure whether the ground value of each pin of J2502 is normal with multimeter diode gear.

4. If the ground value is normal, measure whether the power supply and control signals in the "Display Measurement Table" are normal. If the ground value is normal,

5. Replace U501.

Maintenance Case 1

Fault phenomenon: flower screen

Faulty Component: Software Upgrade

Maintenance mode: flower screen appears during standby, and the fault is repaired after brushing the machine.

Maintenance Case 2

Fault Phenomenon:
Screen Fault
Component: R2539

Maintenance mode: Flash screen, flower screen, visual inspection R2539 is damaged, and the fault is repaired after replacement.

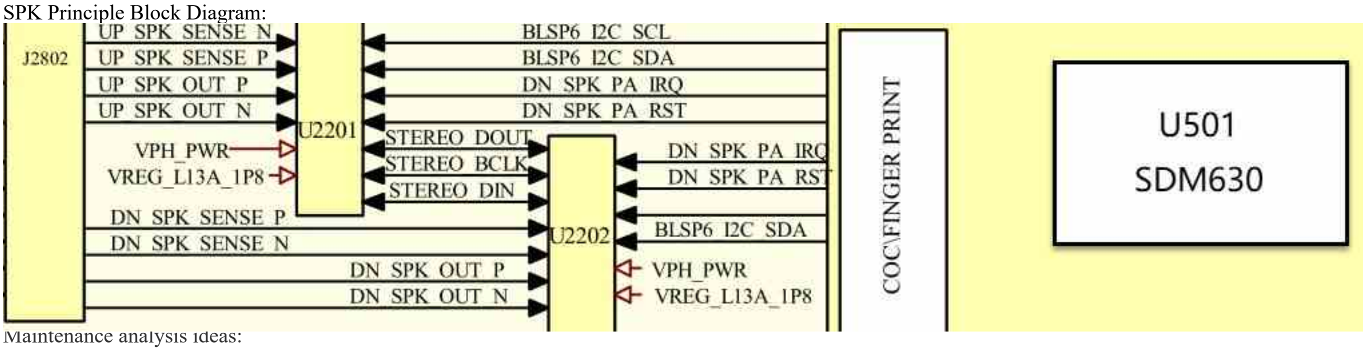
3.8 Audio fault

The audio circuit of Xiaomi Tablet 4 includes: dual speakers, microphones and earphones (excluding earphones). After the audio signals come out of U1601, they go to different channels respectively. First, distinguish which part has the problem according to the fault phenomenon, and then analyze and repair according to the following modules.

3.8.1 Loudspeaker failure

Speaker is connected to the motherboard through FPC. Its principle is to go to CODEC through CPU first, and then amplify and output to interface J2802 through U2201 and U2202 audio power amplifiers and then to speakers.

SPK Principle Block Diagram:



Maintenance analysis ideas:

1 Visually check whether J2802 looks good.

2. Software upgrade troubleshooting software.

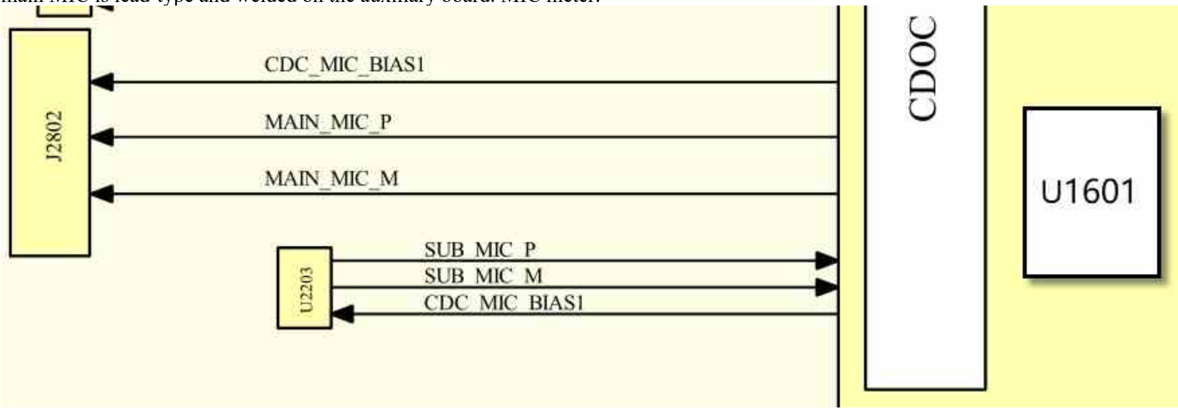
3. Make sure that one of the two speakers is faulty.

4. Measure whether the working conditions of U2201 and U2202 are normal.

4. If the above signals are normal, consider whether the bus from CODEC to CPU is normal.

3.8.2 MIC fault

The main board of Xiaomi Plate 4 includes two MIC circuits, main/auxiliary MIC, and the main MIC is lead-type and welded on the auxiliary board. MIC meter:



Maintenance analysis ideas:

1. Visually check whether J2802 looks good.
2. Measure whether MAIN_MIC_P, MAIN_MIC_M and CDC_MIC_BIAS1 are normal to ground.
3. Measure whether the CDC_MIC_BIAS1 voltage is normal.
4. If the above signals are normal, consider whether the bus between CODEC and CPU is normal.

3.8.3 Headphone failure

Maintenance analysis ideas:

1. Check whether the earphone interface contacts are deformed and whether there are foreign bodies in the earphone interface.
2. Software upgrade and troubleshooting.
3. Measure whether the earphone interface is normal to ground, do not insert the earphone when starting up, and measure whether the DETECT detection signal has 1.8 V voltage.
4. Repair the headphone channel according to the principle block diagram, and replace U1601 normally.
5. Replace U1601.

Schematic diagram of earphone interface:

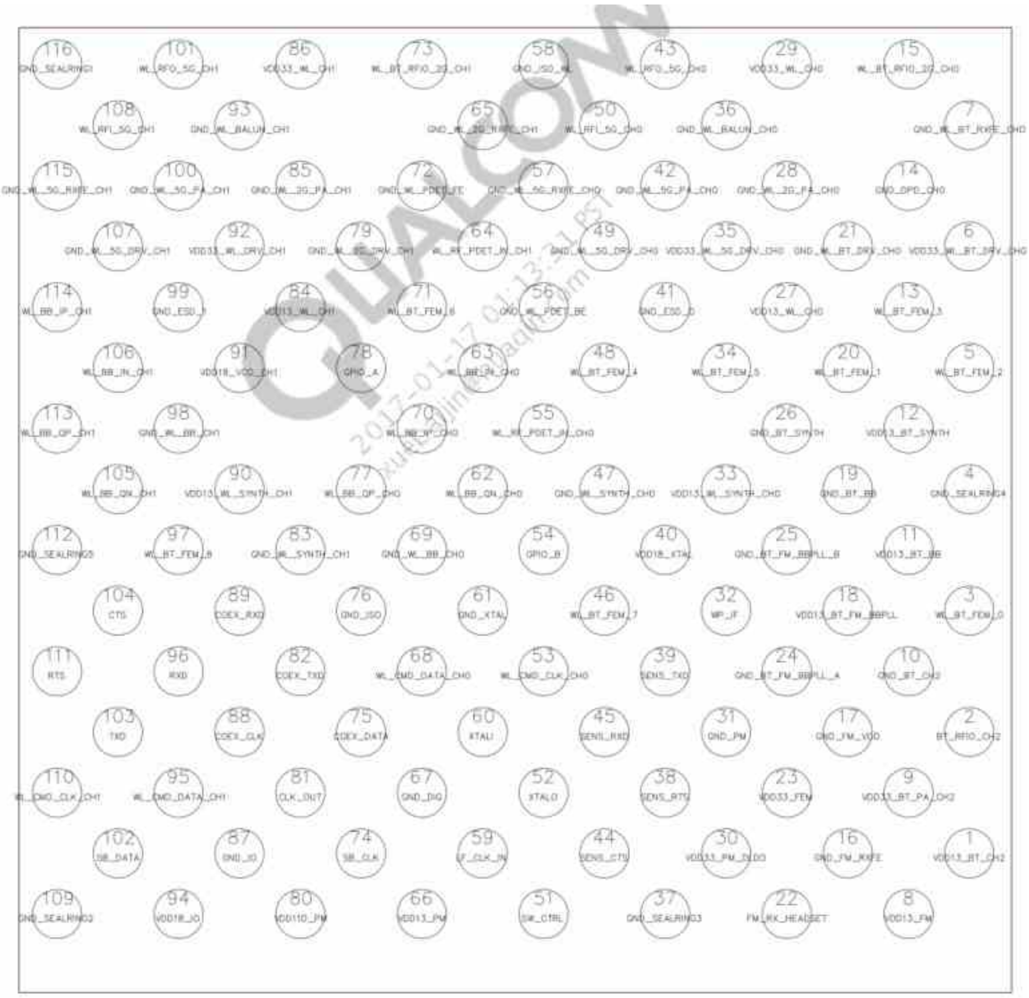
Maintenance Case 1

Fault phenomenon: Silent fault component of earphone left channel: U1201

Maintenance mode: The left channel of earphone is silent, the resistance to ground of B2305 is abnormal, and the fault is repaired after replacing U1201.

3.9 WIFI/BT/FM/GPS Fault

U4001 Bitmap:



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U4001电压测量表		
Symbol	测量值	测量点
VREG_L6A_1P3	1.3V	R4016
REG_L13A_1P8	1.8V	C4015
VREG_L9A_1P8	1.2V	C4012
VDD11D_PM	1V	C4005
VREG_L19A_3P3	3.3V	R4015
VREG_L8B_3P3	3.3V	C4032
SLEEP_CLK	32KHz	C3007

ew)

Maintenance analysis ideas:

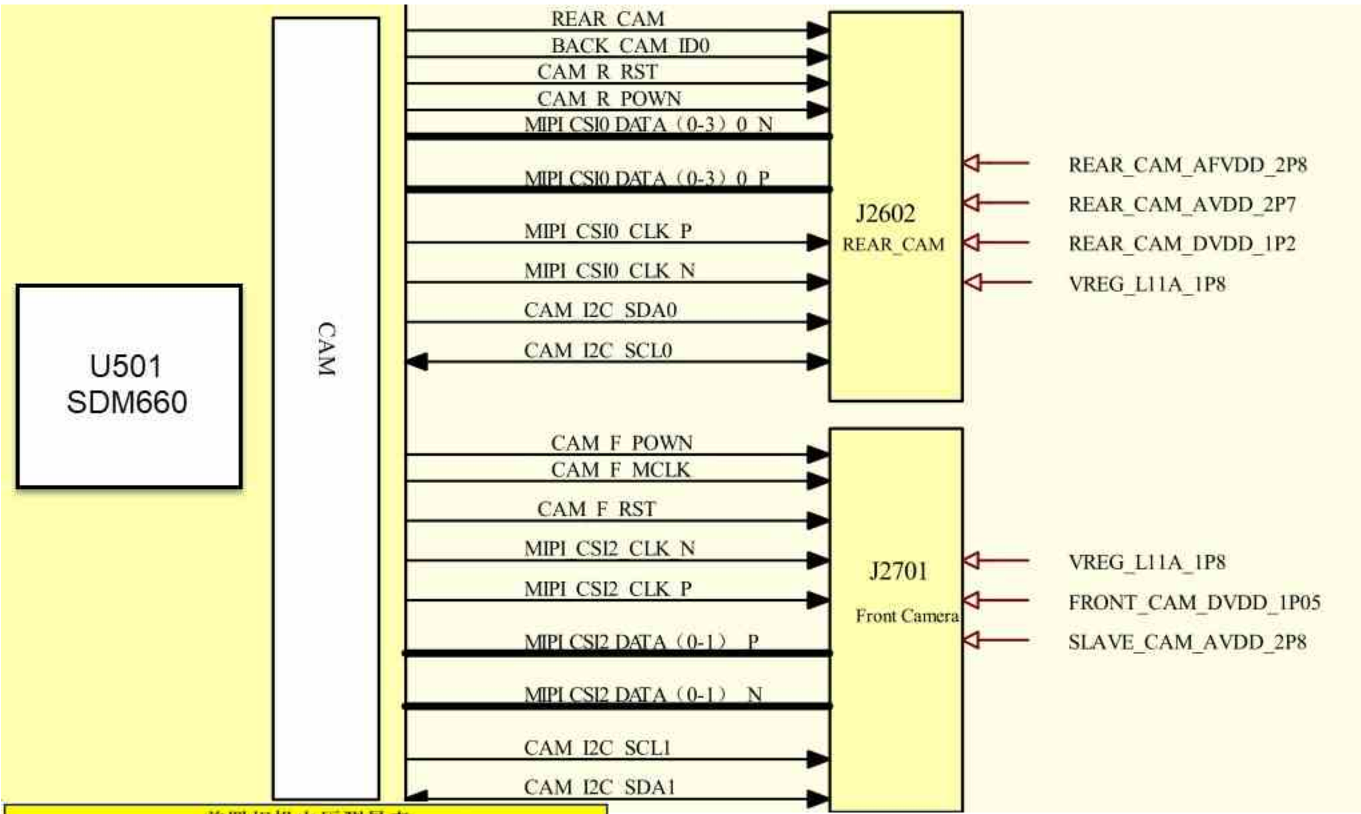
1. Software upgrade, troubleshooting software.
2. Measure whether X4001 is normal (48MHz).
3. Measure whether the power supply, clock and enable signals of U4000 are normal.
4. Take off U4001 and measure whether the bus between U2001 is normal. If it is normal, replace U4001.
5. Replace U501.

Maintenance Case 1
Failure phenomenon:
WiFi Internet access is
slow. Failure reason:
Software upgrade

Maintenance method: WiFi is slow to surf the Internet, and the fault is repaired after brushing the machine.

3.10 Camera failure

Principle block diagram:



前置相机电压测量表		
Symbol	测量值	测量点
SLAVE_CAM_AVDD_2P8	2.8V	C2705
FRONT_CAM_DVDD_1P05	1.05V	C2707
CAM_I2C_SCL1	1.8V	R807
CAM_I2C_SDA1	1.8V	R806
CAM_F_RST	1.8V	C2702
CAM_F_MCLK	1V	R2701
MIPI_CSI2_CLK_P/N	760	Z2705
MIPI_CSI2_DATA0-1 P/N	760	Z2703/Z2704

Rear camera signal measurement:

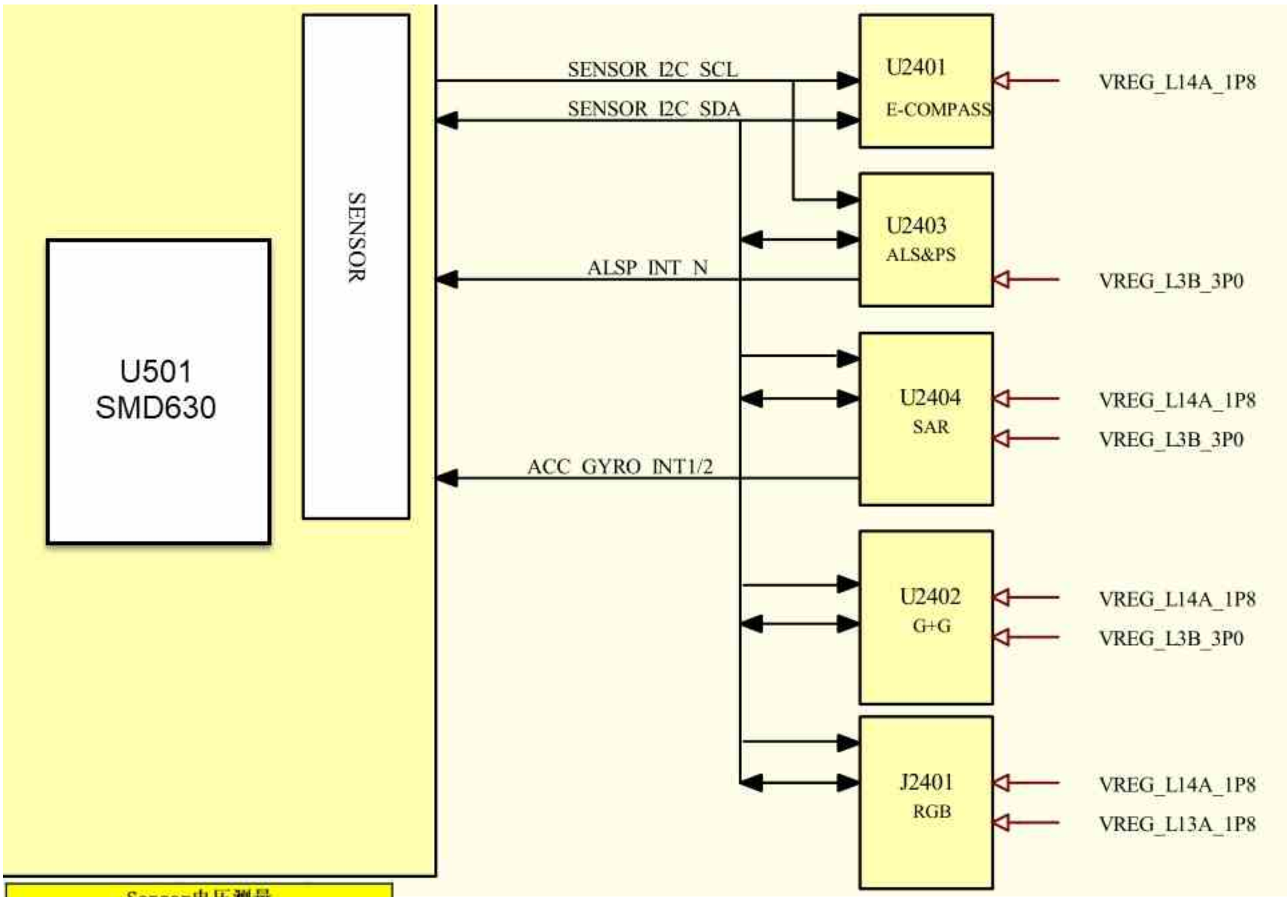
后置相机电压测量表		
Symbol	测量值	测量点
REAR_CAM_AFVDD_2P8	2.8V	C2712
REAR_CAM_AVDD_2P7	2.7V	R2608
VREG_L11A_1P8	1.8V	R2610
REAR_CAM_DVDD_1P2	1.2V	C2614
CAM_I2C_SDA0	1.8V	R805
CAM_I2C_SCL0	1.8V	R804
CAM_R_RST	1.8V	C2605
CAM_R_MCLK	1V	R2602
MIPI_CSI0_DATA0-3_N/P	760	Z260（2/4/6/8）
MIPI_CSI0_CLK_P/N	760	Z2610

1. Software upgrade, troubleshooting software.
2. Detect J2602, J2701 and surrounding components for loss and damage.
3. Go to CIT to test the front camera and the rear camera to distinguish faults.
4. Measure whether the ground values of J2602 and J2701 are normal.
5. Measure whether the camera power supply, clock and reset signal output are normal.
6. Measure whether the I2C and MIPI bus output by U501 are normal.

Maintenance Case 1
Failure Phenomenon:
Faulty components can't
be opened after shooting:
Software upgrade
Maintenance mode: The back camera can't be opened, and the fault can be repaired after brushing the machine.

3.11 Inductor failure

Principle block diagram:



Sensor电压测量		
Symbol	测量值	测量点
SENSOR_I2C_SCL	1.8V	R810
SENSOR_I2C_SDA	1.8V	R811
VREG_L14A_1P8	1.8V	C1416
VREG_L3B_3P0	3V	C1710
VREG_L13A_1P8	1.8V	C2590

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.

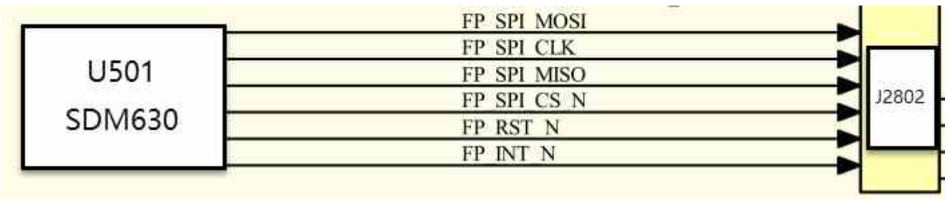
2. Measure whether the working conditions of the corresponding sensors are normal.

3. Replace the corresponding sensor.

4. Replace U501.

3.12 Fingerprint identification failure

Principle block diagram:



Fingerprint voltage meter:

指纹电压测量表		
Symbol	测量值	测量点
FP_VDDIO	1.8V	C2805
FP_AVDD	2.8V	C2806
FP_SPI_MISO	1.8V	J1900第23PIN
FP_RST_N	1.8V	C2807

fingerprint interface and surrounding components are damaged, and if there is damage, please replace it.

2. Brush the machine to eliminate software faults.

3. Measure whether the above voltage and other signals are normal.

4. If all the above signals are normal, replace U501.